■ Outer

`Outer[`f`, ` $list_1$`, ` $list_2$`, ` ...`]` gives the generalized outer product of the $list_i$.

Example: `Outer[f,{a,b},{x,y}]` ⟶ `{{f[a, x], f[a, y]}, {f[b, x], f[b, y]}}`.
■ `Outer[Times, ` $list_1$`, ` $list_2$`, ` ...`]` gives an outer product. ■ The result of applying `Outer` to two tensors $T_{i_1 i_2 \ldots i_r}$ and $U_{j_1 j_2 \ldots j_s}$ is the tensor $V_{i_1 i_2 \ldots i_r j_1 j_2 \ldots j_s}$ with elements `f[`$T_{i_1 i_2 \ldots i_r}$`, ` $U_{j_1 j_2 \ldots j_s}$`]`. Applying `Outer` to two tensors of ranks r and s gives a tensor of rank $r + s$. ■ The heads of all the $list_i$ must be the same, but need not necessarily be `List`. ■ The $list_i$ need not necessarily be cuboidal arrays. ■ See page 457. ■ See also: `Inner`, `Distribute`.

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